

From Redundant Tendon Tensions to Low-Dimensional Actuation Coordinates: A SVD-Based Method for Tendon-Actuated Continuum Robots

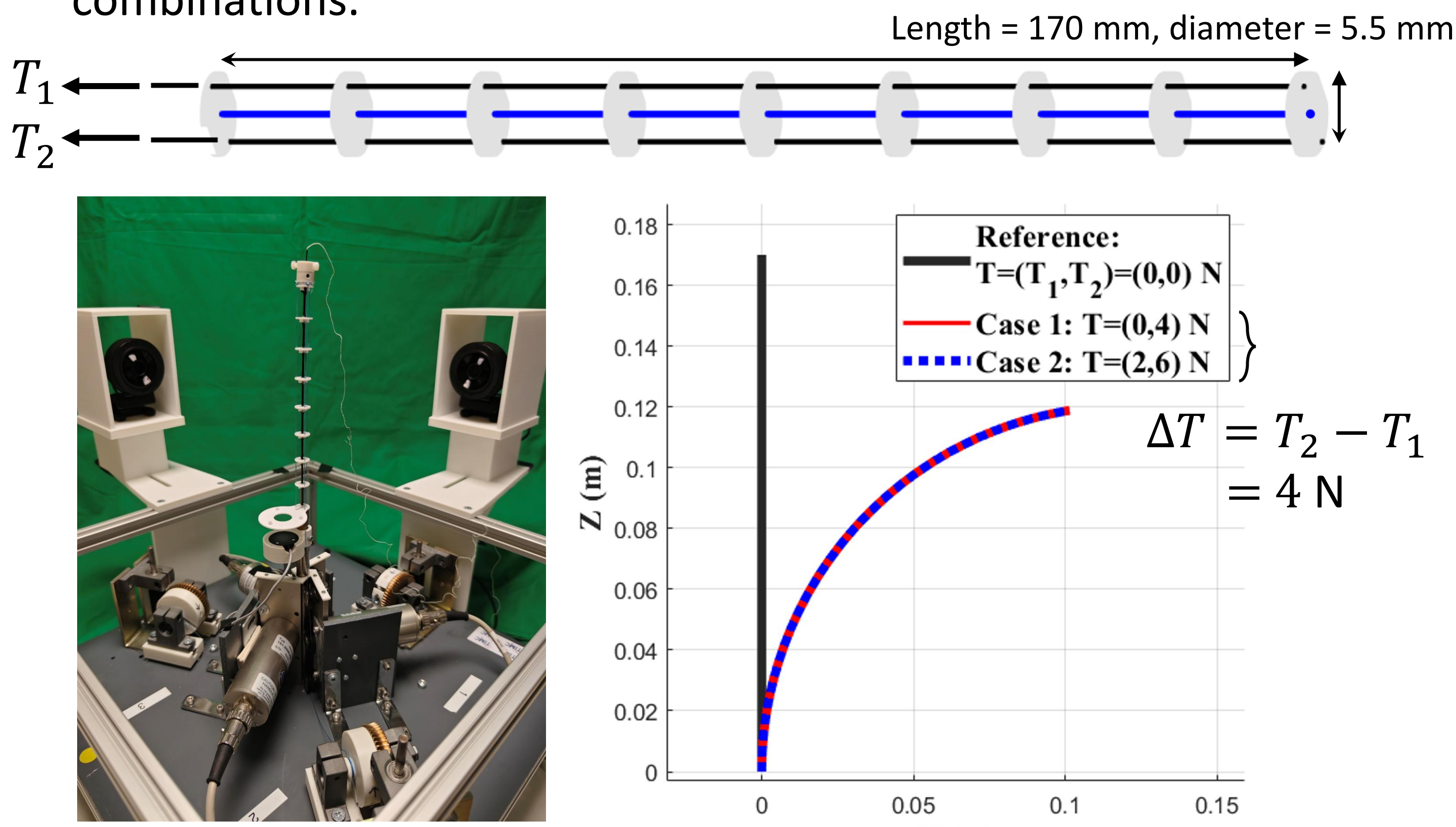
Zibo Zhang^{1,2}, M. Taha Chikhaoui², Vincent Lebastard¹, and Frédéric Boyer¹

¹LS2N Laboratory, Institut Mines Telecom Atlantique Atlantique, 44307 Nantes, France,

²Univ. Grenoble Alpes, CNRS, UMR 5525, VetAgro Sup, Grenoble INP, TIMC, 38000 Grenoble, France

Motivation & Problem

- Tendon-Actuated Continuum Robots (TACRs) are often over-actuated.
- The same shape can be produced by multiple tendon tension combinations.



- As a result, predicting the tendon actuation for a given shape becomes challenging.
- Tracking successive desired shapes (shape control) also becomes challenging, especially for multi-routing and multi-segment TACRs.

Key Contributions

- Rank characterization of actuation redundancy
- Generic SVD-based method for actuation redundancy reduction
- Construction of reduced equivalent routings
 - Expressed in a low-dimensional actuation space
 - Preserving the same deformation as the original routing
 - Using fewer actuation inputs and having lower computational cost than original redundant routing
- Comparison with Clarke analytical reduction method [2]

Static Model of A TACR

$$Q(q) + Kq = L(q)T \quad [1]$$

with $Q(q)$: configuration-dependent internal generalized forces, K : generalized stiffness matrix, q : generalized deformation coordinates, $L(q)$: actuation matrix, T : tendon tension vector

Characterization of Actuation Redundancy

- Active-passive strain decomposition:

$$\epsilon = \epsilon_a + \epsilon_p = \Phi_a q_a + \Phi_p q_p$$

ϵ : strain field, Φ : strain function basis, q : generalized coordinates of deformation, $(\cdot)_a$ and $(\cdot)_p$: associated to actuated and passive strain modes, respectively.

- Using the routing-dependent strain basis Φ_{aT} :

$$q_a = S(\Phi_{aT}) T$$

- Redundancy is detected through rank deficiency of S . If $\text{rank}(S) < m$ (number of tendons), then:

$$T = T_{eff} + z, \quad z \in \text{Ker}(S)$$

T_{eff} contributes to deformation while z does not.

Redundancy Reduction Method

Clarke-based method [2]

Projection of T onto a two-dimensional actuation space:

$$u = M_p T$$

With the projection matrix M_p :

SVD-based method

$$Q(q) + Kq = L(q)T$$

$$\downarrow$$

$$Q(q) + Kq = L_0 T$$

$$\downarrow$$

$$L_0 = U \Sigma V^T$$

$$\downarrow$$

Clarke-based method

$$M_p = \frac{2}{m} \begin{bmatrix} \cos(\psi_1) & \cos(\psi_2) & \dots & \cos(\psi_m) \\ \sin(\psi_1) & \sin(\psi_2) & \dots & \sin(\psi_m) \end{bmatrix}$$

and angular positions of tendons in cross-section frame:

$$\psi_i = \frac{2\pi}{m} (i - 1), i = 1, \dots, m$$

Both u and τ_a are low-dimensional actuation coordinates.

SVD-based method

$$(\Phi_a, \Phi_p), \tau_a$$

$$\downarrow$$

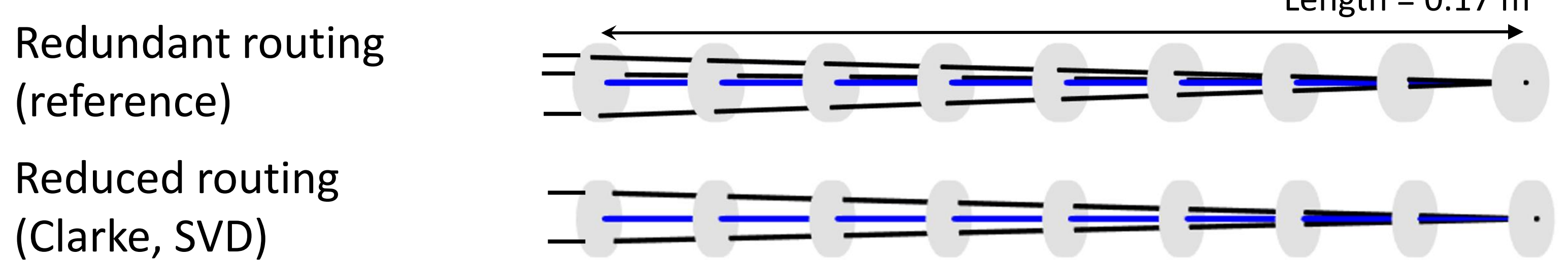
Segregated static model expressed in low-dimensional actuation space:

$$\begin{pmatrix} Q_a \\ Q_p \end{pmatrix} + \begin{pmatrix} K_{aa} & K_{ap} \\ K_{pa} & K_{pp} \end{pmatrix} \begin{pmatrix} q_a \\ q_p \end{pmatrix} = \begin{pmatrix} L_a \tau_a \\ 0 \end{pmatrix}$$

L_a, τ_a : reduced actuation matrix and coordinates

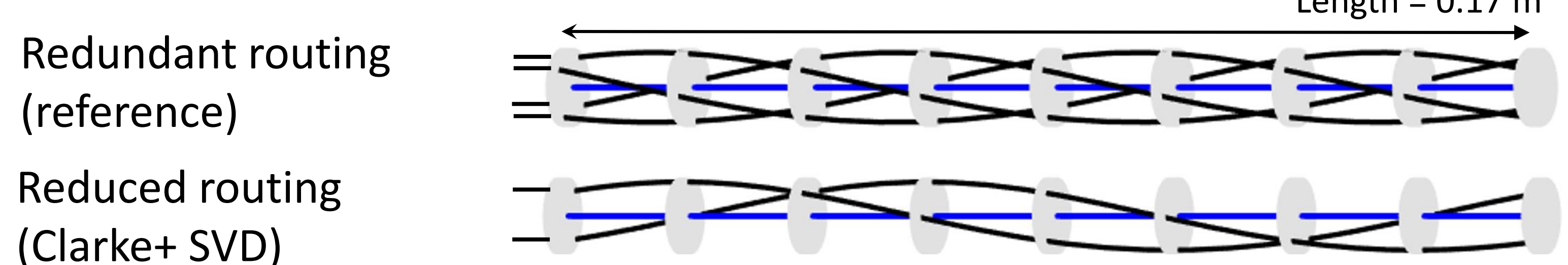
Application to TACR Redundant Routings

Example 1: single-segment with 3 convergent tendons



Routing	Actuation inputs	Tip error	RMS	Max
Redundant	3 physical tensions	Clarke, SVD	0.0014 %	0.0024 %
Reduced	Clarke, SVD: 2 bending coordinates	Both Clarke and SVD work for symmetric planar routings.		

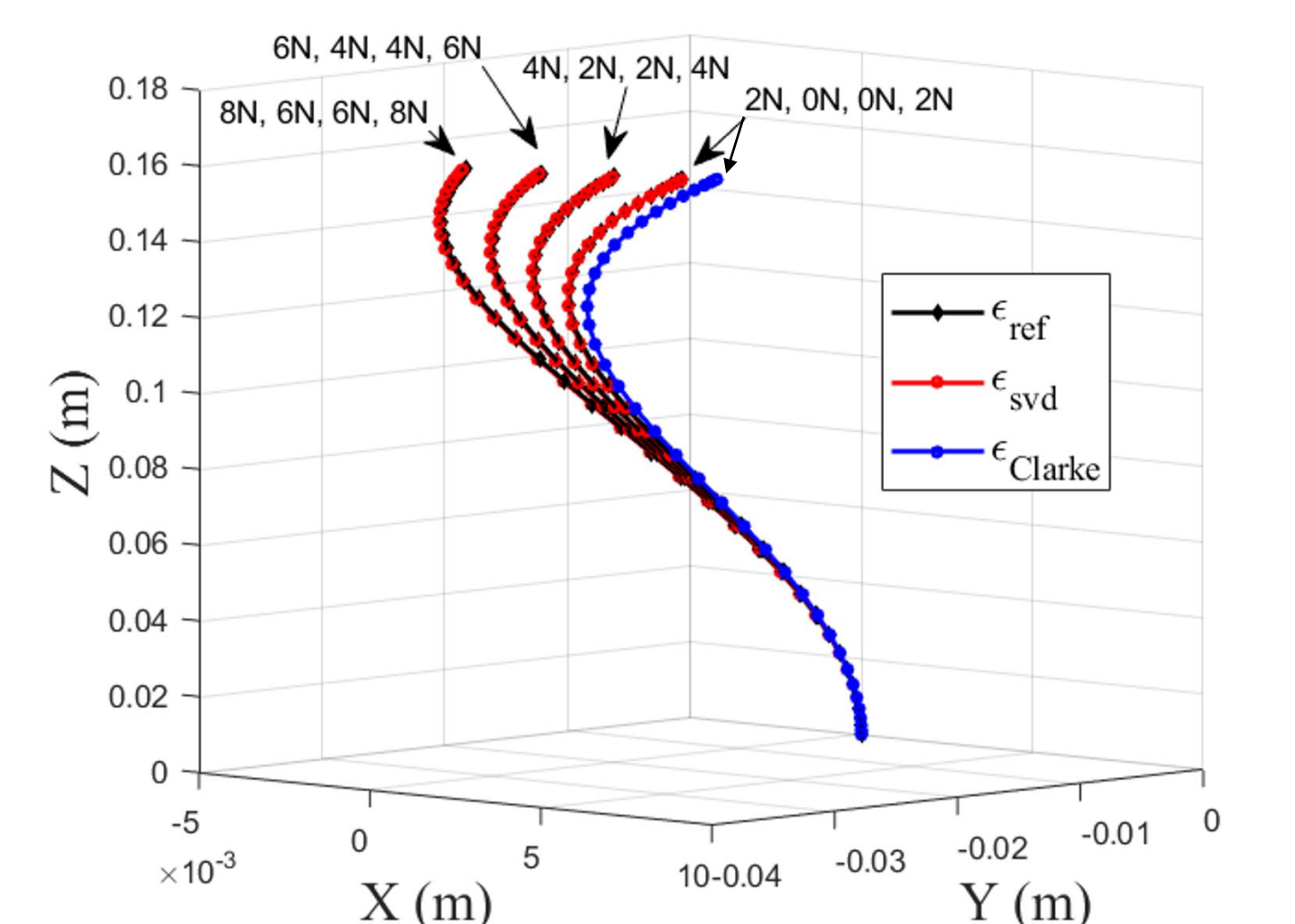
Example 2: single-segment with 4 spiral tendons



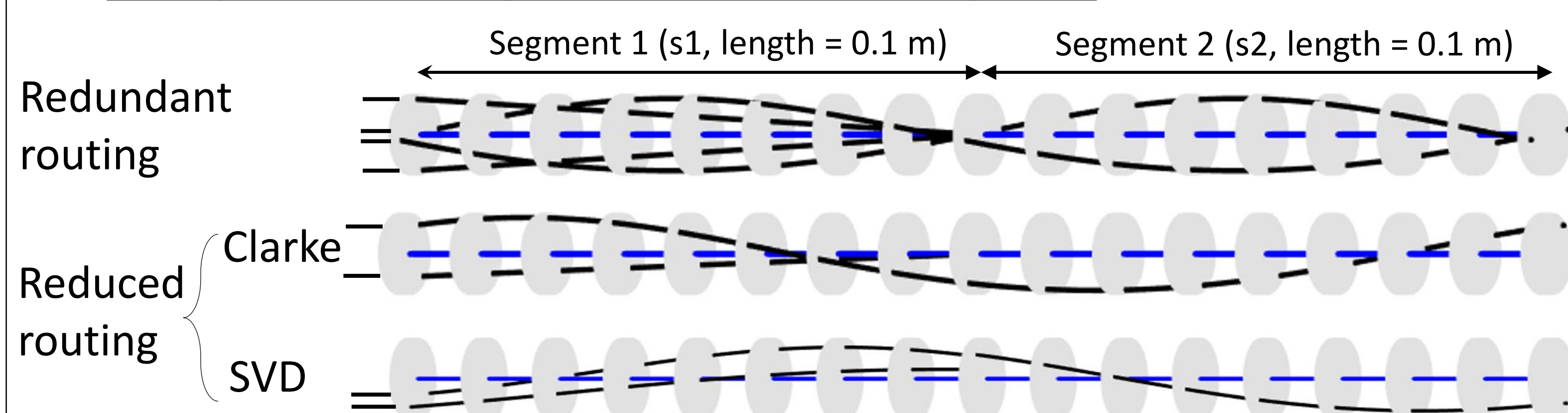
Routing	Actuation inputs
Redundant	4 physical tensions
Reduced	Clarke: 2 bending SVD: 2 bending + 1 torsion

Tip error	RMS	Max
Clarke	6.16 %	22.33 %
SVD	0.21 %	0.36 %

Clarke fails for torsion (3D deformation); SVD works for both 2D and 3D.



Example 3: multi-segment and multi-routing TACR



Routing	Actuation inputs of s1	Actuation inputs of s2
Redundant	4 tendon tensions	2 tendon tensions
Reduced	Clarke: 2 bending SVD: 2 bending + 1 torsion	Clarke: 1 bending SVD: 1 bending + 1 torsion

Tip error	RMS	Max
Clarke	0.99 %	2.87 %
SVD	0.08 %	0.24 %

SVD provides a general reduction valid for 2D, 3D, and multi-segments, while Clarke is limited to symmetric planar routings.

Computational gain of SVD-based reduced routing: **~5x** faster than the redundant routing

Future Work

- Develop a shape control law based on the virtual reduced routing and the segregated static model.
- Validate the developed shape control law experimentally.

This work is supported by the French National Agency for Research under awards No. ANR-20-CE33-0001, ANR-21-ESRE-0015, and ANR-24-CE33-4611.

[1] F. Boyer et al., *IEEE Transactions on Robotics*, 2021.

[2] R. Grassmann and J. Burgner-Kahrs, *IEEE Robotics and Automation Letters*, 2025.